# EE2211 Tutorial 6

(Ridge Regression in Dual Form)
**Question 1:**

Derive the solution for linear ridge regression in dual form (see Lecture 6 notes page 16).

**Answer:** For $\lambda > 0$,

$$(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

$$\Rightarrow \mathbf{X}^T\mathbf{X}\mathbf{w} + \lambda\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

$$\Rightarrow \lambda\mathbf{w} = \mathbf{X}^T\mathbf{y} - \mathbf{X}^T\mathbf{X}\mathbf{w}$$

$$\Rightarrow \mathbf{w} = \lambda^{-1}(\mathbf{X}^T\mathbf{y} - \mathbf{X}^T\mathbf{X}\mathbf{w})$$

$$\Rightarrow \mathbf{w} = \lambda^{-1}\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$\mathbf{w} = \mathbf{X}^T\boldsymbol{a}$$

where

$$\boldsymbol{a} = \lambda^{-1}(\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$\Rightarrow \lambda\boldsymbol{a} = (\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$\Rightarrow \lambda\boldsymbol{a} = (\mathbf{y} - \mathbf{X}\mathbf{X}^T\boldsymbol{a})$$

$$\Rightarrow \mathbf{X}\mathbf{X}^T\boldsymbol{a} + \lambda\boldsymbol{a} = \mathbf{y}$$

$$\Rightarrow (\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})\boldsymbol{a} = \mathbf{y}$$

$$\Rightarrow \boldsymbol{a} = (\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})^{-1}\mathbf{y}$$

Hence,

$$\mathbf{w} = \mathbf{X}^T\boldsymbol{a} = \mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})^{-1}\mathbf{y}.$$

(Polynomial Regression, 1D data)
**Question 2:**

Given the following data pairs for training:

$\{x = -10\} \rightarrow \{y = 5\}$
$\{x = -8\} \rightarrow \{y = 5\}$
$\{x = -3\} \rightarrow \{y = 4\}$
$\{x = -1\} \rightarrow \{y = 3\}$
$\{x = 2\} \rightarrow \{y = 2\}$
$\{x = 8\} \rightarrow \{y = 2\}$

(a) Perform a 3rd-order polynomial regression and sketch the result of line fitting.
(b) Given a test point $\{x = 9\}$ predict $y$ using the polynomial model.
(c) Compare this prediction with that of a linear regression.

**Answer:**

Polynomial model of 3rd order: $f(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_1^2 + w_3 x_1^3$.

$$\mathbf{P} = \begin{bmatrix} 1 & -10 & 100 & -1000 \\ 1 & -8 & 64 & -512 \\ 1 & -3 & 9 & -27 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \\ 1 & 8 & 64 & 512 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}.$$

Polynomial regression:

$$\hat{\mathbf{w}} = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{y}$$

$$= \begin{bmatrix} 6 & -12 & 242 & -1020 \\ -12 & 242 & -1020 & 18290 \\ 242 & -1020 & 18290 & -100212 \\ -1020 & 18290 & -100212 & 1525082 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ -10 & -8 & -3 & -1 & 2 & 8 \\ 100 & 64 & 9 & 1 & 4 & 64 \\ -1000 & -512 & -27 & -1 & 8 & 512 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2.6894 \\ -0.3772 \\ 0.0134 \\ 0.0029 \end{bmatrix}$$

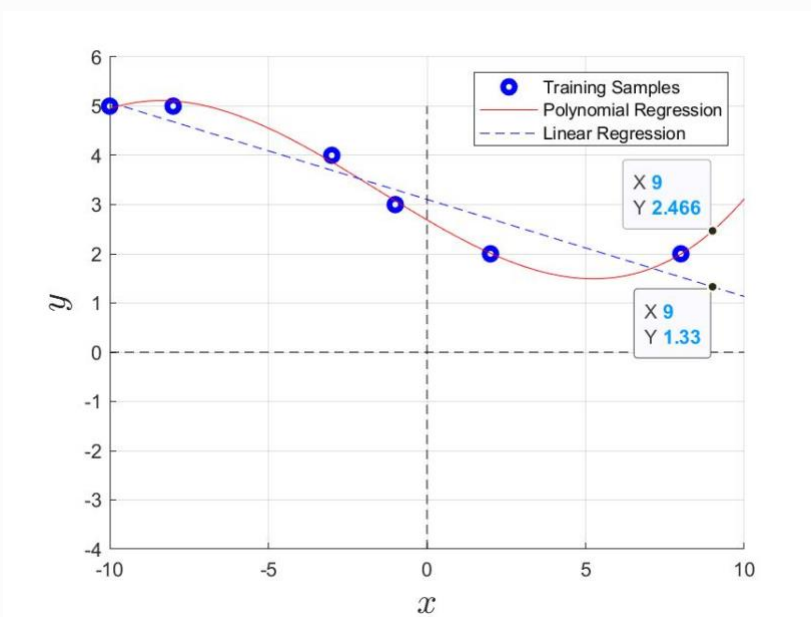
Linear regression:

$$\hat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 6 & -12 \\ -12 & 242 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ -10 & -8 & -3 & -1 & 2 & 8 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3.1055 \\ -0.1972 \end{bmatrix}.$$

Prediction:

y_predict_Poly = 2.4661

y_predict_Linear = 1.3303



(Polynomial Regression, 3D data, Python)

**Question 3:**

(a) Write down the expression for a 3rd order polynomial model having a 3-dimensional input.

(b) Write down the **P** matrix for this polynomial given $\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}$.

(c) Given $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, can a unique solution be obtained in dual form? If so, proceed to solve it.

(d) Given $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, can the primal ridge regression be applied to obtain a unique solution? If so, proceed to solve it.

**Answer:**

(a) Polynomial model of 3rd order:

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$
$$+ w_{12}\,x_1x_2 + w_{23}\,x_2x_3 + w_{13}\,x_1x_3 + w_{11}\,x_1^2 + w_{22}\,x_2^2 + w_{33}\,x_3^2$$
$$+ w_{211}\,x_2x_1^2 + w_{311}\,x_3x_1^2 + w_{122}\,x_1x_2^2 + w_{322}\,x_3x_2^2 + w_{133}\,x_1x_3^2 + w_{233}\,x_2x_3^2$$
$$+ w_{123}\,x_1x_2x_3 + w_{111}\,x_1^3 + w_{222}\,x_2^3 + w_{333}\,x_3^3 \quad \text{_____(1)}$$

(b) **P** = [

Columns 1 through 13

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | -1 | 1 | -1 | -1 | 1 | 1 | 1 | 1 | -1 | 1 | 1 |

Columns 14 through 20

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | -1 | -1 | 1 | -1 | 1 ] |

(c) Yes

$$\hat{\mathbf{w}} = \mathbf{P}^T(\mathbf{P}\mathbf{P}^T)^{-1}\mathbf{y} = \mathbf{P}^T \begin{bmatrix} 10 & 10 \\ 10 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\hat{\mathbf{w}}^T$ =[0 0 -0.1000 0 -0.1000 -0.1000 0 0 0.1000 0 -0.1000
0 0.1000 0.1000 0 -0.1000 -0.1000 0 -0.1000 0]

In python:

```
w_dual =
[ 0.  0.  -0.1  0.  0. -0.1  0.  0.1 -0.1  0.  0.  -0.1  0.  0.1  -0.1  0. -0.1  0.1 -0.1  0. ]
```

(Note: The arrangement of the polynomial terms in the columns of matrix **P** using `PolynomialFeatures` from `sklearn.preprocessing` might be different from that in equation(1).)

(d) Yes

$$\hat{\mathbf{w}} = (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{y}$$

$\hat{\mathbf{w}}^T$ =[0.0000 0.0000 -0.1000 0.0000 -0.1000 -0.1000 0.0000
0.0000 0.1000 0.0000 -0.1000 0.0000 0.1000 0.1000
0.0000 -0.1000 -0.1000 0.0000 -0.1000 0.0000]

In python:

```
w_primal = [ 9.99969302e-07  9.99972940e-07 -9.99980001e-02  9.99970098e-07
```

```
 9.99970666e-07 -9.99980000e-02  9.99967597e-07  9.99980000e-02
-9.99980000e-02  9.99972485e-07  9.99969529e-07 -9.99980000e-02
 9.99968506e-07  9.99980000e-02 -9.99980001e-02  9.99970553e-07
-9.99980001e-02  9.99980001e-02 -9.99980001e-02  9.99969416e-07]
```

(Note: The arrangement of the polynomial terms in the columns of matrix **P** using `PolynomialFeatures` from `sklearn.preprocessing` might be different from that in equation(1).)

Here, at $\lambda = 0.0001$, we observe a very close solution to that in (c) even though (d) constitutes an **approximation** whereas (c) is exact.

**Codes:**

```
import numpy as np
from numpy.linalg import inv
from sklearn.preprocessing import PolynomialFeatures
X = np.array([[1,0,1], [1,-1,1]])
y = np.array([0, 1])
## Generate polynomial features
order = 3
poly = PolynomialFeatures(order)
P = poly.fit_transform(X)
## dual solution (without ridge)
w_dual = P.T @ inv(P @ P.T) @ y
print(w_dual)
## primal ridge
reg_L = 0.0001*np.identity(P.shape[1])
w_primal_ridge = inv(P.T @ P + reg_L) @ P.T @ y
print(w_primal_ridge)
```

(Binary Classification, Python)

**Question 4:**

Given the training data:

$$\begin{aligned}
\{x = -1\} &\rightarrow \{\, y = class1 \,\} \\
\{x = \ \ 0\,\} &\rightarrow \{\, y = class1 \,\} \\
\{x = 0.5\} &\rightarrow \{\, y = class2 \,\} \\
\{x = 0.3\} &\rightarrow \{\, y = class1 \,\} \\
\{x = 0.8\} &\rightarrow \{\, y = class2 \,\}
\end{aligned}$$

Predict the class label for $\{x = -0.1\}$ and $\{x = 0.4\}$ using linear regression with signum discrimination.

**Answer:**

$$\mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 0.5 \\ 1 & 0.3 \\ 1 & 0.8 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} +1 \\ +1 \\ -1 \\ +1 \\ -1 \end{bmatrix}$$

$$\hat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \begin{bmatrix} 0.3333 \\ -1.1111 \end{bmatrix}$$

$$\mathrm{sgn}(\hat{\mathbf{y}}_t) = \mathrm{sgn}(\mathbf{X}_t\hat{\mathbf{w}}) = \mathrm{sgn}\left(\begin{bmatrix} 0.4444 \\ -0.1111 \end{bmatrix}\right) = \begin{bmatrix} class + 1 \\ class - 1 \end{bmatrix} \begin{matrix} \rightarrow class1 \\ \rightarrow class2 \end{matrix}$$

**Codes:**

```
import numpy as np
from numpy.linalg import inv
from sklearn.preprocessing import PolynomialFeatures
```

```
X = np.array([[1,-1], [1,0], [1,0.5], [1,0.3], [1,0.8]])
y = np.array([1, 1, -1, 1, -1])
## Linear regression for classification
w = inv(X.T @ X) @ X.T @ y
print(w)
Xt = np.array([[1,-0.1], [1,0.4]])
y_predict = Xt @ w
print(y_predict)
y_class_predict = np.sign(y_predict)
print(y_class_predict)
```

(Multi-Category Classification, Python)

**Question 5:**

Given the training data:

$$\begin{aligned}
\{x = -1\} &\rightarrow \{\, y = class1 \,\} \\
\{x = \ \ 0\} &\rightarrow \{\, y = class1 \,\} \\
\{x = 0.5\} &\rightarrow \{\, y = class2 \,\} \\
\{x = 0.3\} &\rightarrow \{\, y = class3 \,\} \\
\{x = 0.8\} &\rightarrow \{\, y = class2 \,\}
\end{aligned}$$

(a) Predict the class label for $\{x = -0.1\}$ and $\{x = 0.4\}$ based on linear regression towards a one-hot encoded target.

(b) Predict the class label for $\{x = -0.1\}$ and $\{x = 0.4\}$ using a polynomial model of 5th order and a one-hot encoded target.

**Answer:**

$$\mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 0.5 \\ 1 & 0.3 \\ 1 & 0.8 \end{bmatrix}, \ \mathbf{Y} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \ \mathbf{X}_{\mathrm{t}} = \begin{bmatrix} 1 & -0.1 \\ 1 & 0.4 \end{bmatrix}.$$

(a) $\widehat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} = \begin{bmatrix} 0.4780 & 0.3333 & 0.1887 \\ -0.6499 & 0.5556 & 0.0943 \end{bmatrix}$

$$\widehat{\mathbf{Y}}_{\mathrm{t}} = \mathbf{X}_{\mathrm{t}}\widehat{\mathbf{w}} = \begin{bmatrix} 1 & -0.1 \\ 1 & 0.4 \end{bmatrix} \begin{bmatrix} 0.4780 & 0.3333 & 0.1887 \\ -0.6499 & 0.5556 & 0.0943 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5430 & 0.2778 & 0.1792 \\ 0.2180 & 0.5556 & 0.2264 \end{bmatrix} \Rightarrow \begin{bmatrix} class1 \\ class2 \end{bmatrix}$$

(b) Polynomial model of 5th order: $f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_1^2 + w_3x_1^3 + w_4x_1^4 + w_5x_1^5$

$$\mathbf{P} = \begin{bmatrix} 1.0000 & -1.0000 & 1.0000 & -1.0000 & 1.0000 & -1.0000 \\ 1.0000 & 0 & 0 & 0 & 0 & 0 \\ 1.0000 & 0.5000 & 0.2500 & 0.1250 & 0.0625 & 0.0313 \\ 1.0000 & 0.3000 & 0.0900 & 0.0270 & 0.0081 & 0.0024 \\ 1.0000 & 0.8000 & 0.6400 & 0.5120 & 0.4096 & 0.3277 \end{bmatrix}.$$

$$\widehat{\mathbf{w}} = \mathbf{P}^T(\mathbf{P}\mathbf{P}^T)^{-1}\mathbf{Y} = \begin{bmatrix} 1.0000 & 0 & -0.0000 \\ -5.3031 & -3.7023 & 9.0055 \\ 5.2198 & 10.8728 & -16.0926 \\ 6.6662 & 9.4698 & -16.1360 \\ -6.4765 & -12.9099 & 19.3864 \\ -2.6199 & -7.8045 & 10.4244 \end{bmatrix}.$$

$$\mathbf{P}_t = \begin{bmatrix} 1.0000 & -0.1000 & 0.0100 & -0.0010 & 0.0001 & -0.0000 \\ 1.0000 & 0.4000 & 0.1600 & 0.0640 & 0.0256 & 0.0102 \end{bmatrix}.$$

$$\hat{\mathbf{Y}}_t = \mathbf{P}_t\hat{\mathbf{w}} = \begin{bmatrix} 1.5752 & 0.4683 & -1.0435 \\ -0.0521 & 0.4544 & 0.5977 \end{bmatrix} \Rightarrow \begin{bmatrix} class1 \\ class3 \end{bmatrix}.$$

**Codes:**

```
import numpy as np
from numpy.linalg import inv
from sklearn.preprocessing import PolynomialFeatures
X = np.array([[1,-1], [1,0], [1,0.5], [1,0.3], [1,0.8]])
Y = np.array([[1,0,0], [1,0,0], [0,1,0], [0,0,1], [0,1,0]])

## Linear regression for classification
W = inv(X.T @ X) @ X.T @ Y
print(W)
Xt = np.array([[1,-0.1], [1,0.4]])
y_predict = Xt @ W
print(y_predict)
y_class_predict = [[1 if y == max(x) else 0 for y in x] for x in y_predict ]
print(y_class_predict)

## Polynomial regression for
## Generate polynomial features
order = 5
poly = PolynomialFeatures(order)
## only the data column (2nd) is needed for generation of polynomial terms
reshaped = X[:,1].reshape(len(X[:,1]),1)
P = poly.fit_transform(reshaped)
reshaped = Xt[:,1].reshape(len(Xt[:,1]),1)
Pt = poly.fit_transform(reshaped)
## dual solution (without ridge)
Wp_dual = P.T @ inv(P @ P.T) @ Y
print(Wp_dual)
yp_predict = Pt @ Wp_dual
print(yp_predict)
yp_class_predict = [[1 if y == max(x) else 0 for y in x] for x in yp_predict ]
print(yp_class_predict)
```

(Multi-Category Classification, Python)

**Question 6 (continued from Q3 of Tutorial 2):**

Get the data set "`from sklearn.datasets import load_iris`". Use Python to perform the following tasks.

(a) Split the database into two sets: 74% of samples for training, and 26% of samples for testing. Hint: you might want to utilize `from sklearn.model_selection import train_test_split` for the splitting.

(b) Construct the target output using one-hot encoding.

(c) Perform a linear regression for classification (without inclusion of ridge, utilizing one-hot encoding for the learning target) and compute the number of test samples that are classified correctly.

(d) Using the same training and test sets as in above, perform a 2nd order polynomial regression for classification (again, without inclusion of ridge, utilizing one-hot encoding for the learning target) and compute the number of test samples that are classified correctly. Hint: you might want to use `from sklearn.preprocessing import PolynomialFeatures` for generation of the polynomial matrix.

**Codes:**

```
## (a) split data
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
iris_dataset['data'], iris_dataset['target'], test_size=0.26, random_state=0)

## (b) one-hot encoding
# Ytr_onehot = list()
# for i in y_train:
#     letter = [0, 0, 0]
#     letter[i] = 1
#     Ytr_onehot.append(letter)
# Yts_onehot = list()
# for i in y_test:
#     letter = [0, 0, 0]
#     letter[i] = 1
#     Yts_onehot.append(letter)
from sklearn.preprocessing import OneHotEncoder
onehot_encoder=OneHotEncoder(sparse=False)
reshaped = y_train.reshape(len(y_train), 1)
Ytr_onehot = onehot_encoder.fit_transform(reshaped)
reshaped = y_test.reshape(len(y_test), 1)
Yts_onehot = onehot_encoder.fit_transform(reshaped)

## (c) Linear Classification

bias1 = np.ones((X_train.shape[0], 1))
X_train = np.concatenate((bias, X_train), axis = 1)
Bias2 = np.ones((X_test.shape[0], 1))
X_test = np.concatenate((bias, X_test), axis = 1)

w = inv(X_train.T @ X_train) @ X_train.T @ Ytr_onehot
print(w)
# calculate the output based on the estimated w and test input X and then assign
to one of the classes based on one hot encoding
yt_est = X_test.dot(w);
yt_cls = [[1 if y == max(x) else 0 for y in x] for x in yt_est ]
print(yt_cls)
# compare the predicted y with the ground truth
m1 = np.matrix(Yts_onehot)
m2 = np.matrix(yt_cls)
difference = np.abs(m1 - m2)
print(difference)
# calculate the error rate/accuracy
correct = np.where(~difference.any(axis=1))[0]
accuracy = len(correct)/len(difference)
print(len(correct))
print(accuracy)

## (d) Polynomial Classification
import numpy as np
from sklearn.preprocessing import PolynomialFeatures
poly = PolynomialFeatures(2)
P = poly.fit_transform(X_train)
Pt = poly.fit_transform(X_test)
if P.shape[0] > P.shape[1]:
   wp = inv(P.T @ P) @ P.T @ Ytr_onehot
```

```
else:
   wp = P.T @ inv(P @ P.T) @ Ytr_onehot
print(wp)
yt_est_p = Pt.dot(wp);
yt_cls_p = [[1 if y == max(x) else 0 for y in x] for x in yt_est_p ]
print(yt_cls_p)
m1 = np.matrix(Yts_onehot)
m2 = np.matrix(yt_cls_p)
difference = np.abs(m1 - m2)
print(difference)
correct_p = np.where(~difference.any(axis=1))[0]
accuracy_p = len(correct_p)/len(difference)
print(len(correct_p))
print(accuracy_p)
```

(c) Correct prediction 28/39

(d) Correct prediction 38/39

**Question 7**

MCQ: there could be more than one answer. Given three samples of two-dimensional data points $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 3 & 3 \end{bmatrix}$ with corresponding target vector $\mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Suppose you want to use a full third-order polynomial model to fit these data. Which of the following is/are true?

a) The polynomials model has 10 parameters to learn
b) The polynomial learning system is an under-determined one
c) The learning of the polynomial model has infinite number of solutions
d) The input matrix **X** has linearly dependent samples
e) None of the above

**Answer**: a, b, c, d

**Question 8**

MCQ: there could be more than one answer. Which of the following is/are true?

a) The polynomial model can be used to solve problems with nonlinear decision boundary.

b) The ridge regression cannot be applied to multi-target regression.

c) The solution for learning feature $\mathbf{X}$ with target $\mathbf{y}$ based on linear ridge regression can be written as $\hat{\mathbf{w}} = (\mathbf{X}^T\mathbf{X}+\lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$ for $\lambda > 0$. As $\lambda$ increases, $\hat{\mathbf{w}}^T\hat{\mathbf{w}}$ decreases.

d) If there are four data samples with two input features each, the full second-order polynomial model is an over-determined system.

**Answer**: a, c